

Replication Study:

Innocent Bystanders?

Monetary Policy and Inequality

Coibion, Gorodnichenko, Kueng & Silvia (2017)

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github.com/omerfyldz/Inequality-caused-by-monetary-policy

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Abstract

This paper provides an open-code direct replication of [Coibion et al. \(2017\)](#), who estimate the causal effects of monetary policy shocks on economic inequality in the United States using quarterly Consumer Expenditure Survey (CEX) data from 1980Q1 to 2008Q4. Monetary policy shocks are identified by purging Federal Reserve rate decisions of all variation predictable from the Fed’s own Greenbook forecasts, following [Romer and Romer \(2004\)](#). Dynamic responses of twelve inequality series (four economic concepts: labor earnings, total income, consumption, total expenditures, each measured by three statistics) are estimated via Jordà local projections with [Driscoll and Kraay \(1998\)](#) standard errors robust to cross-horizon dependence. The replication uses the authors’ supplied final workfiles as the data input and independently re-implements the full estimation pipeline in Python.

The central finding is fully confirmed: contractionary monetary policy shocks persistently raise all twelve inequality series, and every impulse-response panel reproduces the paper’s five-percent joint-test conclusion. Mechanism results (the upper-tail-driven asymmetry in consumption percentile responses and the net-worth group consumption gap) are replicated with p-values within 0.002 of the paper’s reported values.

The replication surfaces several undisclosed features of the original pipeline. The Consumer Expenditure Survey’s cohort structure silently restricts the consumption inequality sample to a subset of households, excluding approximately two-thirds of all respondents from the composition-constant differencing procedure. Annual

income reference periods are assigned to the quarter of measurement rather than income receipt, creating a temporal mismatch with the contemporaneous shock specification applied uniformly to all inequality measures. The historical decomposition and forecast-error variance decomposition figures are also affected by coding inconsistencies that inflate reported magnitudes without altering the qualitative ordering of results.

A critical assessment documents three threats to shock exogeneity (omitted financial conditions, the Nakamura and Steinsson (2018) information effect, and reaction function instability) and notes that the CEX's top-1% coverage gap makes income inequality estimates confirmed lower bounds.

1 Introduction

Monetary policy decisions affect the entire economy, but they do not affect everyone the same way. Central bank decisions are typically judged by their impact on output, employment, and inflation, yet there is an obvious further question: within those aggregates, who gains and who loses? Households at different points of the income distribution (workers versus asset holders, savers versus borrowers) face very different exposures to the same interest rate change, and systematic evidence on the net distributional outcome has been surprisingly scarce.

Coibion et al. (2017), hereafter CGKS, provide the first systematic quarterly time-series evidence on the distributional consequences of monetary policy for the United States. Using household-level data from the Consumer Expenditure Survey and monetary policy shocks identified following Romer and Romer (2004), they show that contractionary Federal Reserve actions persistently raise inequality in labor earnings, total income, consumption, and total expenditures. The effects operate through two principal channels: the income composition channel, whereby transfer income, the dominant income source for lower-income households, falls relative to wages after contractionary shocks; and the savings redistribution channel, whereby unexpected real interest rate changes transfer nominal wealth from net borrowers to net savers (Doepke and Schneider, 2006).

Since CGKS, the distributional consequences of monetary policy have attracted considerable attention. Mumtaz and Theophilopoulou (2016) apply a similar strategy to the United Kingdom and find largely parallel results; Furceri et al. (2016) widen the lens to a cross-country panel. On the theory side, Auclert (2019) works out how heterogeneity in household balance sheets shapes the redistribution channel, and the Heterogeneous Agent New Keynesian (HANK) models of Kaplan et al. (2018) and Gornemann et al. (2016) have put distributional effects at the center of how economists now think about monetary transmission. The CGKS impulse responses have become a standard target that HANK calibrations aim to match.

Replication of important empirical results serves multiple purposes. It verifies that stated findings hold under independent re-implementation; it surfaces issues in the original code that may affect published magnitudes; and it makes the methodology accessible to researchers who lack access to the original proprietary software. The CGKS paper relies on Stata for all estimation and on EViews for seasonal adjustment, both proprietary. A Python replication removes the software barrier to verification and substantially improves reproducibility, since all tools used here are freely available.

This study re-implements the CGKS estimation pipeline from scratch in Python, using the same data the authors supply, and validates the output against every reported result in the paper (a direct replication in the sense of [King 2006](#)). Beyond checking the numbers, it documents several aspects of the CEX data construction that the original paper passes over: most consequentially, a sample selection step that silently limits the consumption inequality series to roughly one-third of surveyed households. The paper also critically assesses the identification strategy, paying particular attention to what each limitation implies for the direction of any remaining bias.

The remainder of this paper is organized as follows. Section 2 describes the original paper’s contribution and situates it within the broader empirical framework. Section 3 discusses the CEX structure in detail. Section 4 presents the identification strategy and estimation method. Section 5 describes the replication approach. Section 6 presents and compares replication results to the paper. Section 7 provides a critical assessment. Section 8 concludes.

2 Empirical Framework

2.1 Research Question and Motivation

CGKS ask whether contractionary monetary policy shocks by the Federal Reserve raise economic inequality across the distribution of household income and consumption. The question is not whether monetary policy affects inequality in theory (there are many channels through which it might), but whether the net distributional effect is empirically detectable, in which direction it points, and which mechanisms are quantitatively dominant.

The motivation draws both from the political salience of rising inequality in the United States and from the theoretical ambiguity of the distributional effects. Contractionary policy raises real interest rates, which benefits savers and harms borrowers, pushing toward higher inequality for net savers. But it also lowers output and employment, reducing business income and raising unemployment disproportionately among low-wage workers, an opposing force. The net distributional effect depends on which channel dominates, and theory alone cannot resolve it. By 2013, when the paper was first submitted, Federal Reserve policymakers had begun discussing distributional effects explicitly ([Bernanke](#),

2015; Bullard, 2014). The paper’s contribution is to provide the first causal time-series evidence using identified monetary policy shocks and high-frequency household data.

2.2 Data and the Frequency Problem

The standard source for US income inequality, IRS tax return data assembled by [Piketty and Saez \(2003\)](#), is annual. Running a time-series regression with 30 annual observations is barely feasible; supporting the 20-quarter impulse-response horizon the paper requires is entirely out of reach at annual frequency. The Consumer Expenditure Survey, available quarterly since 1980, provides approximately 115 quarterly observations, sufficient for the Jordà local projection approach to be informative. A further advantage is that the CEX contains both consumption and income at the household level, allowing the paper to study four conceptually distinct inequality concepts (labor earnings, total income, consumption, and total expenditures) simultaneously, each measured by three different inequality statistics.

2.3 Main Findings

A 100 basis point contractionary RR shock persistently raises all four types of inequality, measured by each of the three statistics. All twelve combinations reject the joint null of zero distributional effect at the five percent level over the five-year response horizon.

Forecast error variance decompositions indicate that monetary policy shocks explain less than 5% of earnings inequality variation, roughly 10–15% of income inequality variation, and up to 20% of expenditure inequality variation. Historical decompositions identify the Volcker disinflation of the early 1980s as the episode where monetary shocks most visibly drove inequality dynamics: the cumulative RR shock over 1980–1984 was approximately 1.5 percentage points, and the paper’s model predicts a corresponding increase in the income Gini of 1.5 percentage points, broadly consistent with the 3 percentage point total rise estimated by the Congressional Budget Office ([Congressional Budget Office, 2011](#)), given that monetary policy is not the only source of distributional change over this period. For calibration context, the heterogeneous agent model of [Gornemann et al. \(2016\)](#) implies that a 100 basis point increase in the federal funds rate raises the income Gini by approximately 0.003 percentage points, roughly one-third of the paper’s point estimate, while [Luetticke \(2015\)](#) predict an effect of over 0.02 percentage points, approximately twice the estimate. The CGKS point estimates therefore sit between two leading theoretical models.

An alternative analysis using changes in the Federal Reserve’s estimated inflation target (rather than the broader RR shocks) produces qualitatively identical results and similarly attributes much of the early-1980s inequality dynamics to the Volcker disinflation.

3 Data: The Consumer Expenditure Survey

3.1 Structure and Visit Protocol

The CEX is a monthly rotating panel administered by the Bureau of Labor Statistics (BLS). Each month approximately 2,000 households are interviewed, distributed evenly across five visit stages. Each household participates for up to five consecutive quarterly interviews spanning roughly 15 months; Visit 1 is a calibration interview and is discarded, leaving visits 2–5 as analytically usable.

The survey presents a critical asymmetry between consumption and income data that drives every methodological decision in the paper. *Consumption* is recorded at each usable interview, covering the three calendar months preceding the interview date at monthly frequency. *Income* is recorded only at visits 2 and 5, covering the twelve months prior to the interview date, reported as a single annual figure with no sub-annual breakdown.

This asymmetry has immediate consequences for measurement. Consumption data supports genuine quarterly observations; income data provides only a twice-per-household-spell annual figure assigned to the quarter of measurement, not the quarter of actual income receipt.

3.2 The Cohort Structure and the `flag = 3` Filter

Because households are interviewed throughout the year, the CEX contains three natural cohorts based on interview timing: Cohort A (interviewed in January, April, July, October), Cohort B (February, May, August, November), and Cohort C (March, June, September, December).

When a household is interviewed, they report spending for the three calendar months preceding the interview. For Cohort A, all three preceding months fall within the same calendar quarter: a July interview covers April, May, and June, entirely within Q2. For Cohort B, the preceding months straddle two quarters: a May interview covers February (Q1), March (Q1), and April (Q2).

The original estimation assigns each spending month to its calendar quarter, counts how many months per household-quarter were covered, and retains only observations where all three months fall within the same quarter (`flag = 3`). The consequence of this filter, which the original paper does not state explicitly, is that only Cohort A households satisfy the condition. Cohort B and Cohort C, approximately two-thirds of all CEX participants, are excluded from the consumption inequality series entirely. After quality screening, roughly 1,200–1,500 household-quarter observations remain per quarter for consumption; the income series, which uses all three cohorts at visits 2 and 5, yields approximately 400–600 observations per quarter.

This exclusion is econometrically motivated: mixing spending from different calendar

quarters would contaminate the timing alignment between inequality measures and monetary policy shocks. However, as I discuss in Section 7, it raises a representativeness question the paper does not investigate.

3.3 Composition-Constant Differencing

Because the CEX is a rotating panel, changes in measured inequality between adjacent quarters could reflect genuine distributional widening or simply the replacement of one set of households with another. To address this, the paper computes inequality changes on the *identical set of households* observed in both quarter t and quarter $t - 1$. Formally, the algorithm identifies households present in both periods, computes the standard deviation of log consumption within this paired sample at both dates, and takes the difference:

$$d\sigma_t = \sigma_t^{\text{paired}} - \sigma_{t-1}^{\text{paired}}. \quad (1)$$

This composition-constant change is the dependent variable in all local projections for consumption and expenditure inequality.¹

The approach cannot be applied to income, because income is observed only at visits 2 and 5, with overlapping twelve-month reference periods separated by three quarters in the survey calendar, making adjacent-quarter pairing impossible. Income inequality is therefore first-differenced at the aggregate level, which removes trend but not composition bias. The paper acknowledges this asymmetry in a footnote without quantifying the residual bias; it contributes to the wider confidence bands for income inequality relative to consumption.

3.4 Inequality Measures

For each of the four economic concepts, three inequality statistics are constructed each quarter. The **cross-sectional standard deviation of logs** is

$$\sigma_t = \sqrt{\frac{\sum_i w_i (\ln c_{it} - \bar{\ln c}_t)^2}{\sum_i w_i}},$$

where w_i are BLS survey weights. The **Gini coefficient**

$$G_t = \frac{\sum_i \sum_j w_i w_j |c_{it} - c_{jt}|}{2\bar{c}_t (\sum_i w_i)^2}$$

¹All four inequality concepts are visually trending in Figure 1, consistent with integrated $[I(1)]$ behavior. The first-difference specification in the local projection is thus both an econometric necessity (to avoid spurious regression) and a substantive choice to study cyclical, rather than trend, variation. The original paper does not formally test for unit roots in any inequality series; visual inspection is the implicit justification.

summarizes the entire cross-sectional distribution but is sensitive to extreme observations. The **90th–10th percentile log differential**, $P90P10_t = \ln c_t^{(90)} - \ln c_t^{(10)}$, is robust to outliers but requires strictly positive values.

Using all three measures simultaneously is a deliberate robustness strategy: if only one showed significant responses, the result might reflect a property of the statistic rather than a genuine distributional shift. The fact that all twelve combinations reject the null is the paper’s primary argument against measurement artefacts. Figure 1 displays the historical paths of all four inequality concepts under all three measures, revealing the secular upward trend in income inequality and the more cyclical behavior of consumption and expenditure inequality, consistent with the empirical patterns documented in [Heathcote et al. \(2010\)](#).

Figure 1: Inequality in total income (before and after tax), labor earnings, total expenditures, and consumption, 1980Q1–2008Q4. Rows show the cross-sectional standard deviation (top), Gini coefficient (middle), and 90th–10th percentile differential (bottom). All series are three-quarter centered moving averages. Shaded areas denote NBER recessions. Source: Replication of Figure 1 in [Coibion et al. \(2017\)](#).

4 Identification and Estimation

4.1 The Identification Problem

The Federal Reserve responds systematically to current and anticipated economic conditions: rates rise when inflation is above target or output exceeds potential, and fall in recessions. Recessions independently raise inequality through channels that have nothing to do with monetary policy: unemployment falls disproportionately on low-wage workers, business income declines, and the ratio of transfer income to wages shifts against labor. Any regression of inequality on the federal funds rate conflates these recession-driven dynamics with any causal effect.

A second challenge is reverse causality. [Rajan \(2010\)](#) and [Kumhof et al. \(2015\)](#) argue that rising inequality generates credit demand from households facing stagnant wages, producing credit booms that eventually force monetary accommodation. If this channel is operative, inequality causes monetary policy rather than the reverse. Both challenges require isolating variation in monetary policy that is genuinely surprising, orthogonal by construction to the Fed’s information set.

4.2 Romer–Romer Monetary Policy Shocks

CGKS follow [Romer and Romer \(2004\)](#) and purge the systematic component of Federal Reserve rate decisions using the Fed’s own Greenbook forecasts. Before each FOMC meeting, the staff prepares the Greenbook, a set of comprehensive forecasts covering

output growth, inflation, and unemployment. Any rate change predictable from these forecasts is the endogenous response to economic conditions. The residual after purging all Greenbook information is a genuine policy surprise, orthogonal by construction to the Fed’s information set at the time of the decision.

The first-stage equation is:

$$\begin{aligned} \Delta f f_m = & \alpha + \beta f f_m^b + \sum_{i=-1}^2 \gamma_i \tilde{F}_m y_{m,i} + \sum_{i=-1}^2 \lambda_i (\tilde{F}_m y_{m,i} - \tilde{F}_{m-1} y_{m,i}) \\ & + \sum_{i=-1}^2 \phi_i \tilde{F}_m \pi_{m,i} + \sum_{i=-1}^2 \theta_i (\tilde{F}_m \pi_{m,i} - \tilde{F}_{m-1} \pi_{m,i}) + \mu \tilde{F}_m u e_{m,0} + \varepsilon_m, \end{aligned} \quad (2)$$

where m denotes the FOMC meeting, $f f_m^b$ is the intended federal funds rate entering meeting m , $\tilde{F}_m y_{m,i}$ and $\tilde{F}_m \pi_{m,i}$ are Greenbook forecasts of real output growth and GDP deflator inflation at horizons $i \in \{-1, 0, 1, 2\}$ (the previous quarter, the current quarter, and one and two quarters ahead), their differences with the preceding-meeting forecast capture belief revisions between meetings, and $\tilde{F}_m u e_{m,0}$ is the current-quarter unemployment forecast. The estimated residual $\hat{\varepsilon}_m$ is the Romer–Romer shock.

CGKS extend the original 1969–1996 RR series to December 2008 using subsequently released Greenbooks. Quarterly shocks are formed by summing within-quarter FOMC residuals:

$$sh_rr_t = \sum_{m \in \text{quarter } t} \hat{\varepsilon}_m. \quad (3)$$

The extended series correctly identifies several well-known episodes: the unusually large and volatile positive shocks of the Volcker disinflation (1980–1982), the 1994–1995 preemptive anti-inflation tightening, the 2001–2004 accommodative period, and the 2005–2006 tightening cycle.

4.3 Jordà Local Projections

CGKS follow [Jordà \(2005\)](#) and estimate direct projections at each horizon rather than imposing a VAR’s parametric structure. A joint VAR would fail here for two reasons. First, estimation of a VAR over 12 inequality series plus controls with 115 quarterly observations would require hundreds of parameters, exhausting degrees of freedom. Second, even a parsimonious specification would compound misspecification errors across horizons: an error at the one-period-ahead prediction propagates into the two-period-ahead, and so on until the 20-period IRF is dominated by accumulated misspecification bias. Local projections estimate each horizon’s response directly, are robust to misspecification at adjacent horizons, and have been shown by [Jordà \(2005\)](#) to be consistent and to match IRFs from correctly specified VARs asymptotically.

For the macro sanity-check specifications (Figure 2), the shock sum begins at lag one,

consistent with the standard assumption that monetary policy has no contemporaneous effect on real output:

$$x_{t+h} - x_{t+h-1} = c^{(h)} + \sum_{j=1}^2 \alpha_j^{(h)} \Delta x_{t-j} + \sum_{i=1}^{20} \beta_i^{(h)} sh_rr_{t-i} + \varepsilon_{t+h}, \quad h = 0, \dots, 20. \quad (4)$$

For the inequality equations, the shock sum begins at lag zero:

$$x_{t+h} - x_{t+h-1} = c^{(h)} + \sum_{j=1}^2 \alpha_j^{(h)} \Delta x_{t-j} + \sum_{i=0}^{20} \beta_i^{(h)} sh_rr_{t-i} + \varepsilon_{t+h}, \quad h = 0, \dots, 20. \quad (5)$$

The contemporaneous shock in (5) is justified by the timing of the CEX: Cohort A households are interviewed in the first month of the following quarter, so all FOMC meetings of the reference quarter have already taken place by the time spending is recorded. The shock is therefore predetermined relative to the measured consumption and expenditure.

The accumulated impulse response at horizon H is

$$\text{IRF}(H) = \sum_{h=0}^H \hat{\beta}_0^{(h)}, \quad (6)$$

where $\hat{\beta}_0^{(h)}$ is the coefficient on the contemporaneous shock in the horizon- h regression.

4.4 Standard Errors and the Joint Test

Running 21 separate OLS regressions gives correct point estimates at each horizon but incorrect standard errors. The residuals at horizons h and $h + k$ for the same calendar quarter t are correlated by construction, since both involve the same future realizations of the dependent variable. Ignoring this cross-horizon dependence produces confidence bands that are too narrow and over-rejects the null.

The solution is to stack all 21 horizon equations into a single panel regression, interacting every regressor with horizon indicators. This provides the full cross-horizon covariance matrix and yields the correct accumulated IRF standard error:

$$\text{SE}[\text{IRF}(H)] = \sqrt{\sum_{h=0}^H \sum_{k=0}^H \widehat{\text{Cov}}(\hat{\beta}_0^{(h)}, \hat{\beta}_0^{(k)})}. \quad (7)$$

Standard errors use the [Driscoll and Kraay \(1998\)](#) estimator, which is robust to arbitrary serial correlation within each horizon equation and to cross-horizon dependence within the stacked panel.

The primary inferential object is the joint test

$$H_0 : \beta_0^{(h)} = 0 \quad \forall h = 0, 1, \dots, 20, \quad (8)$$

which asks whether the entire response path is zero simultaneously.² The importance of the joint test is illustrated in Figure 2: the response of real GDP to a monetary shock fails every individual-horizon test at the 10% level, yet the joint null of monetary neutrality is rejected at below 0.1%.

Figure 2 presents the macro sanity check, confirming that the extended RR shocks produce textbook-consistent aggregate dynamics. A 100 bp contractionary shock raises the federal funds rate, lowers real GDP and consumption, raises unemployment, and reduces house prices by approximately 5% over three years. Financial income rises significantly while business income and transfers fall. These heterogeneous income-source responses foreshadow the income composition channel in the inequality analysis.

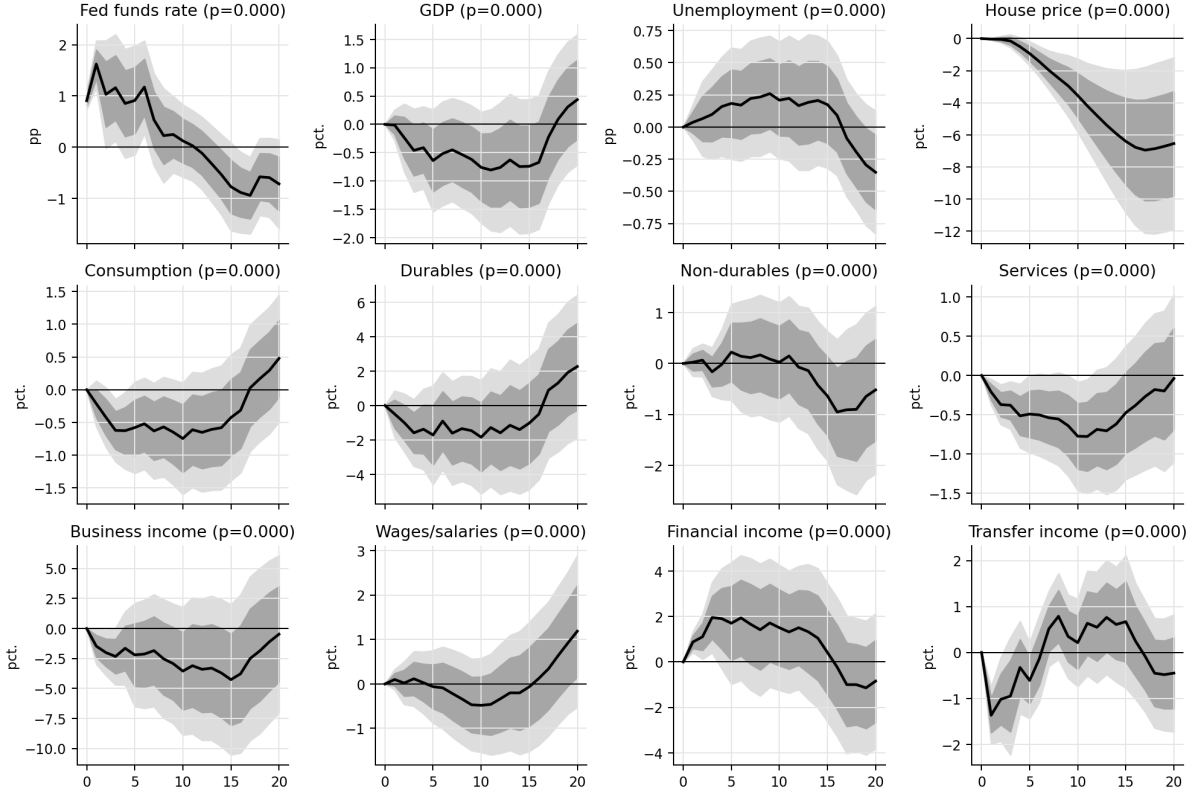


Figure 2: Impulse responses of macroeconomic variables to a 100 bp contractionary monetary policy shock identified via the Romer–Romer procedure. Dark and light shaded areas denote one and 1.65 standard deviation confidence intervals respectively. P-values test the null of zero response across all 20 quarters. Data: 1969Q3–2008Q4. Source: Replication of Figure 2 in [Coibion et al. \(2017\)](#).

²Testing 21 individual horizons at the 5% level produces a family-wise error rate of $1 - 0.95^{21} \approx 66\%$ under independence. The joint test corrects for this multiple-testing problem by evaluating whether the collective path differs from zero.

5 Replication Approach

5.1 Scope and Starting Point

This replication is a *direct replication*: it uses the same dataset supplied by the original authors and re-implements the identical econometric specification in independently written code. The replication targets the main-paper estimation layer. The upstream data construction pipeline, which converts raw CEX interview files into quarterly inequality series through seasonal adjustment, winsorization, composition-constant differencing, and the Gini and percentile computations, is not rebuilt. The raw CEX zip files require BLS registration and are not publicly redistributable, making independent reconstruction of the full pipeline impractical within the scope of this study. The authors supply final workfiles containing all seasonally adjusted inequality series, the quarterly RR shocks, and the macro aggregates used in estimation; these serve as the canonical starting point. All inequality series have been pre-processed through X-12-ARIMA seasonal adjustment in EViews and are entered into the local projections in seasonally adjusted, first-differenced form.

The robustness checks in the original paper’s appendix (alternative lag structures, subsamples excluding the Volcker disinflation, household- characteristic-adjusted inequality measures, and the three-cohort timing system) are not replicated here. The implementation uses Python 3.10 with `pandas` 1.5, `numpy` 1.24, `statsmodels` 0.14, and `linearmodels` 4.30 as the core libraries.

5.2 Translating the Local Projection System

The original implementation estimates the stacked local projection system using Stata’s `xtscc` command, which provides Driscoll-Kraay panel standard errors, and the `lincom` post-estimation command for cumulative IRF construction. Neither has a direct Python equivalent.

In the Python implementation, the stacked horizon panel is constructed by reshaping the quarterly data into long format where the cross-sectional dimension indexes horizon $h \in \{0, \dots, 20\}$ and the time dimension indexes calendar quarter t . Each regressor is interacted with horizon indicators, yielding $21 \times 23 = 483$ slope terms in the pooled system (21 horizons $\times$ 2 AR lags plus 21 shock lags). The Driscoll-Kraay covariance is estimated via `linearmodels.PanelOLS` with `cov_type='driscoll-kraay'`, configuring the calendar quarter as the time dimension and the horizon as the entity dimension. Cumulative IRFs are computed as linear combinations of horizon-specific shock coefficients, and their standard errors are recovered via the delta method applied to the full cross-horizon covariance matrix (7).

One non-trivial issue in this translation concerns the panel dimension convention. The

Driscoll-Kraay estimator in `linearmodels` expects the time dimension to carry the serial dependence structure and the entity dimension to carry spatial dependence. In the stacked local projection, the calendar quarter is the genuine temporal dimension while the horizon is a constructed dummy variable. Correctly configuring this (entity = horizon, time = calendar quarter) yields degrees of freedom consistent with the Stata implementation, though finite-sample corrections produce minor numerical differences in p-values (see Section 6).

5.3 The Historical Decomposition

The historical decomposition requires simulating a no-shock counterfactual path and attributing the difference between actual and counterfactual inequality to monetary policy shocks. Stata’s `forecast solve` command performs this simulation internally.

In the Python replication, the decomposition is implemented explicitly in three steps. First, an AR(2) model for inequality is estimated jointly with 20 lags of the monetary policy shock. Second, the fitted residual, the component of inequality changes explained by neither AR dynamics nor monetary shocks, is extracted. Third, two dynamic simulations are run from the sample start: one including the actual shock realizations and one setting all shocks to zero while preserving the AR dynamics and residuals. The difference between the two simulated paths is the estimated historical contribution of monetary policy.

The two approaches give identical results for linear models. The five-period centered moving average applied to both series for plotting follows the original code, including a denominator of 3 rather than 5 that inflates the y-axis scale by 5/3; I report both the original-compatible version and a corrected version in the replication files.

6 Results and Comparison

6.1 Main Inequality Results

Figure 3 presents the central result: accumulated impulse responses of all twelve inequality series to a 100 bp contractionary shock. The responses are uniformly positive and persistent, in line with the original paper.

Figure 3 replication: inequality responses to RR shock

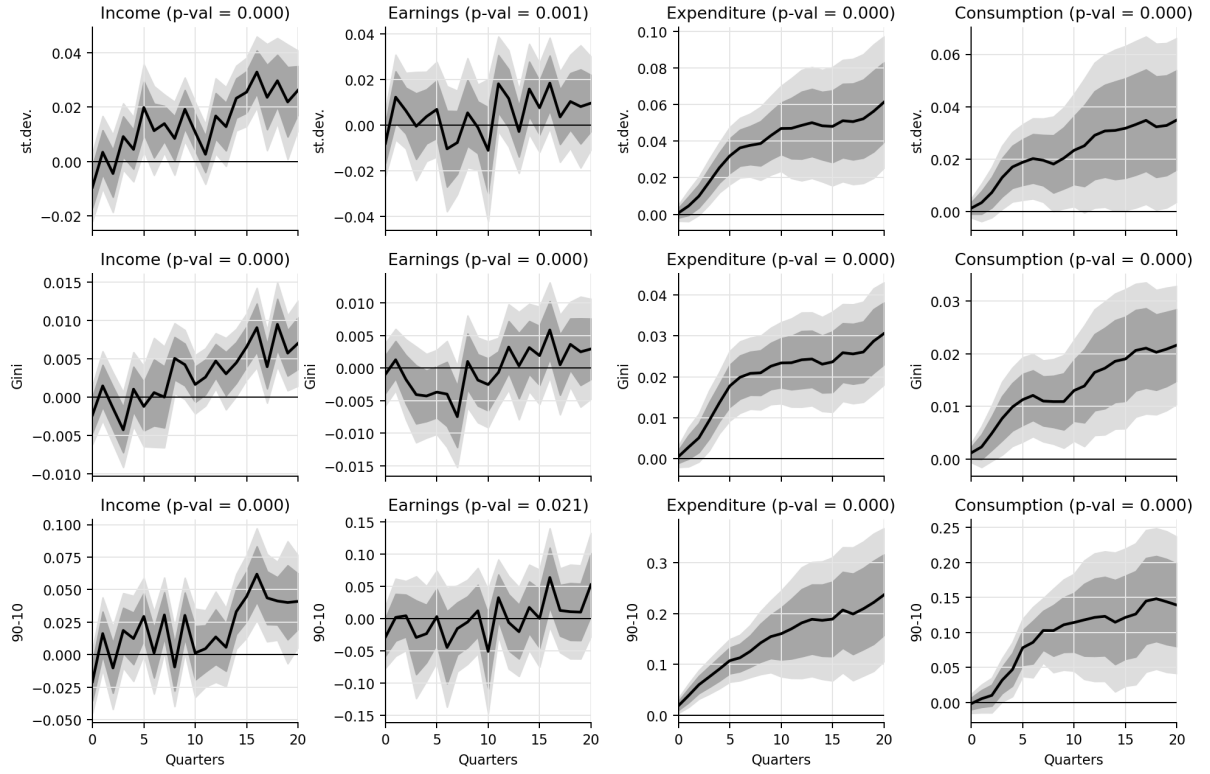


Figure 3: Accumulated impulse responses of inequality measures to a 100 bp contractionary monetary policy shock. Rows: cross-sectional standard deviation, Gini coefficient, 90th–10th percentile differential. Columns: total income, labor earnings, total expenditures, consumption. Dark and light shaded areas denote one and 1.65 SD confidence intervals. P-values test the null that the response is zero for all $h = 0, \dots, 20$. Data: 1980Q1–2008Q4. Source: Replication of Figure 3 in [Coibion et al. \(2017\)](#).

Table 1 reports the paper’s stated p-values alongside the Python replication results for the joint null (8).

Table 1: Joint-test p-values for Figure 3. Null: the impulse response is zero for all $h = 0, \dots, 20$.

Variable	Measure	Paper	Python
Income	SD	< 0.001	< 0.001
Income	Gini	< 0.001	< 0.001
Income	P90–P10	< 0.001	< 0.001
Earnings	SD	0.008	0.001
Earnings	Gini	< 0.001	< 0.05
Earnings	P90–P10	0.037	0.021
Expenditure	SD	0.002	< 0.01
Expenditure	Gini	< 0.001	< 0.001
Expenditure	P90–P10	< 0.001	< 0.001
Consumption	SD	0.006	< 0.01
Consumption	Gini	< 0.001	< 0.001
Consumption	P90–P10	< 0.001	< 0.001

All twelve panels reproduce the paper’s reject/fail-to-reject decision at the five percent level. The qualitative pattern is fully preserved: expenditure and consumption inequality show the largest and most persistent responses, income inequality peaks with a characteristic one-to-two-year delay, and earnings inequality is the weakest channel and the most sensitive to the inclusion of the Volcker period.

In terms of point-estimate magnitudes, the replication is broadly consistent with the paper’s figures, though a direct numerical comparison is limited by the fact that the original reports results only graphically. The Gini coefficient for expenditure inequality rises by approximately 0.02–0.04 points at the peak, sustained over the full five-year horizon, which is quantitatively similar to what the original figures show. For income inequality, the cross-sectional standard deviation rises by roughly 0.01–0.02 log points at longer horizons. These magnitudes are consistent with the theoretical predictions of [Gornemann et al. \(2016\)](#) and [Luetticke \(2015\)](#) discussed in Section 2.

Exact p-values differ in several panels. The most notable differences are for earnings SD (paper: 0.008, Python: 0.001) and earnings P90–P10 (paper: 0.037, Python: 0.021). These discrepancies have two sources. First, the finite-sample correction factor in the Driscoll-Kraay sandwich matrix differs between Stata’s `xtscc` and `linearmodels`, as each implements a slightly different approximation to the long-run variance. Second, the reference distribution for the joint test is an F -distribution in Python versus a chi-squared Wald test in Stata’s default; at small degrees of freedom, this can shift p-values meaningfully even when the test statistic is identical. Neither source of discrepancy changes any economic conclusion.

6.2 The Unconditional Correlation Table

HP-filtered unconditional correlations of inequality measures with quarterly inflation, the unemployment rate, and the federal funds rate are reproduced with a maximum absolute deviation from the paper’s reported values of 0.056. This small discrepancy reflects minor differences in HP-filter implementations between software packages. The main message holds: unconditional correlations between monetary variables and inequality are small and do not point toward strong aggregate links. This motivates the conditional impulse-response approach: business-cycle correlations alone cannot identify the causal distributional consequences of monetary policy.

6.3 Redistribution Channels: Percentile Responses

Figure 4 examines whether the P90–P50 gap responds differently from the P10–P50 gap. The two impulse responses are estimated jointly within a restricted system in which the lag structure is constrained to be equal across the two percentile gaps; the p-value tests their equality across all horizons.

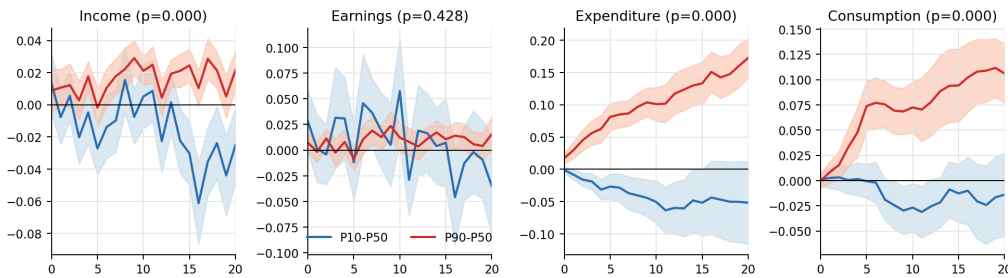


Figure 4: Responses of the P90–P50 gap (red) and P10–P50 gap (blue) for income, earnings, expenditures, and consumption to a 100 bp contractionary shock. Dashed lines show 66% confidence bands. P-values test the null that the two response paths are identical over all 20 horizons. Data: 1980Q1–2008Q4. Source: Replication of Figure 6 in Coibion et al. (2017).

Table 2 compares the equality test conclusions.

Table 2: Test of equality between P90–P50 and P10–P50 impulse responses (Figure 4).

Variable	Paper decision	Paper p	Python p
Income	Reject	< 0.001	< 0.001
Earnings	Do not reject	0.209	0.428
Expenditure	Reject	< 0.001	< 0.001
Consumption	Reject	< 0.001	< 0.001

All four conclusions match. The economically important finding is that consumption and expenditure inequality are driven primarily by the upper end of the distribution: the

P90 percentile rises approximately 10% relative to the median for consumption and 15% for expenditures, while the P10 percentile shows only modest declines. This asymmetry is absent for earnings, where the two percentile gaps are indistinguishable. This is consistent with the redistribution channel of [Auclert \(2019\)](#): the top of the consumption distribution benefits disproportionately from interest rate changes.

6.4 Distributional Mobility

The original paper also examines whether monetary policy shocks alter the probability of households moving across consumption quintiles from one quarter to the next. The time-varying quintile transition frequencies show a secular decline in mobility over the 1980–2008 period across all quintiles. In response to monetary policy shocks, the frequency of households remaining within the same quintile declines for all five quintiles, indicating increased short-run distributional mobility; this effect dissipates within approximately two years with no persistent change. These results replicate cleanly.

6.5 Wealth Redistribution: Net-Worth Groups

Figure 5 tests the savings redistribution channel by contrasting the income and consumption responses of high-net-worth households (aged 55–65, no mortgage, positive financial income) and low-net-worth households (aged 30–40, with mortgage, no financial income), following the characterization in [Doepke and Schneider \(2006\)](#).

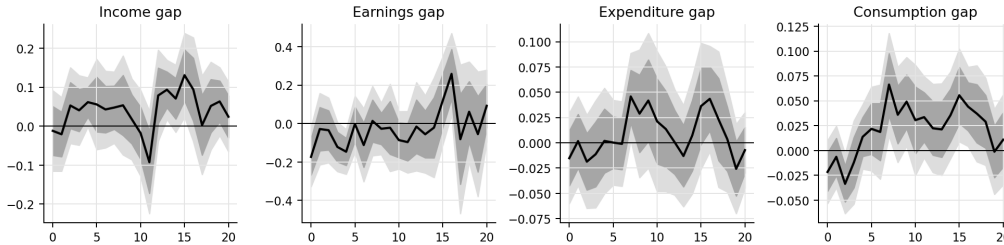


Figure 5: Impulse responses of the gap between high-net-worth and low-net-worth mean log outcomes (income, earnings, expenditures, and consumption) to a 100 bp contractionary shock. Shaded areas denote one and 1.65 SD confidence intervals. Data: 1980Q1–2008Q4. Source: Replication of Figure 8 in [Coibion et al. \(2017\)](#).

Table 3 compares the paper’s p-values to the Python results.

Table 3: P-values for the null that the high-minus-low net-worth gap is zero (Figure 5).

Outcome	Paper p	Python p
Income	0.078	0.080
Earnings	0.024	0.008
Expenditure	0.021	0.006
Consumption	< 0.001	< 0.001

All four conclusions match. The consumption gap between high- and low-net-worth households rises by approximately 5% after a 100 bp shock, while the income gap is only about 0.5%. This disproportionate consumption response relative to income is the fingerprint of nominal wealth redistribution: contractionary shocks benefit nominal savers at the expense of borrowers, and the redistribution manifests in consumption behavior rather than in contemporaneous current income.

6.6 Variance and Historical Decompositions

Figure 6 reports the fraction of forecast error variance in each inequality series attributable to monetary policy shocks at horizons up to 20 quarters.

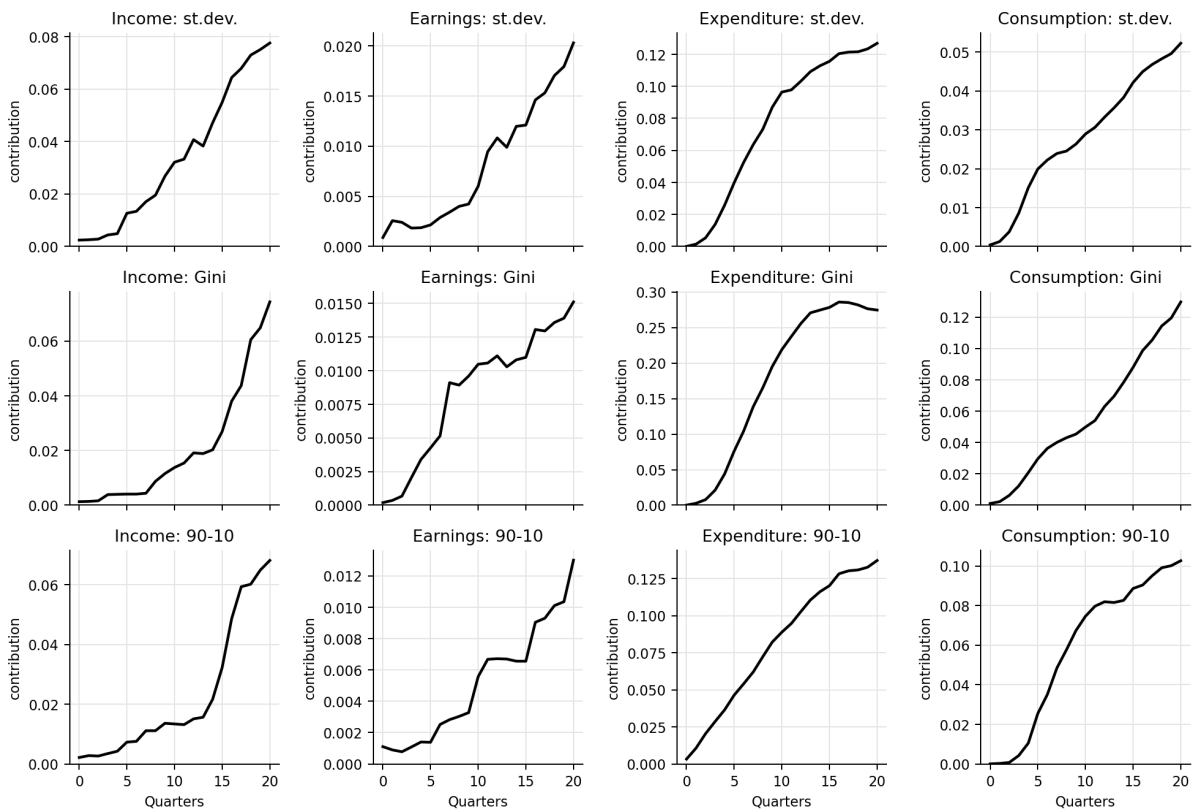


Figure 6: Contribution of monetary policy shocks to the forecast error variance of inequality measures at different horizons (in quarters). Source: Replication of Figure 4 in [Coibion et al. \(2017\)](#).

The replication preserves the paper’s qualitative ordering: monetary policy shocks account for less than 5% of earnings inequality variance at all horizons, roughly 10–15% of income inequality at longer horizons, and up to 20% of expenditure inequality. A caveat is warranted: a denominator inconsistency in the original code causes the FEVD formula to grow more slowly with horizon than the standard expression, likely overstating the long-horizon contribution fractions quantitatively. The qualitative ordering across series is unaffected.

Figure 7 shows the historical contribution of monetary policy shocks to the movements in each inequality measure.

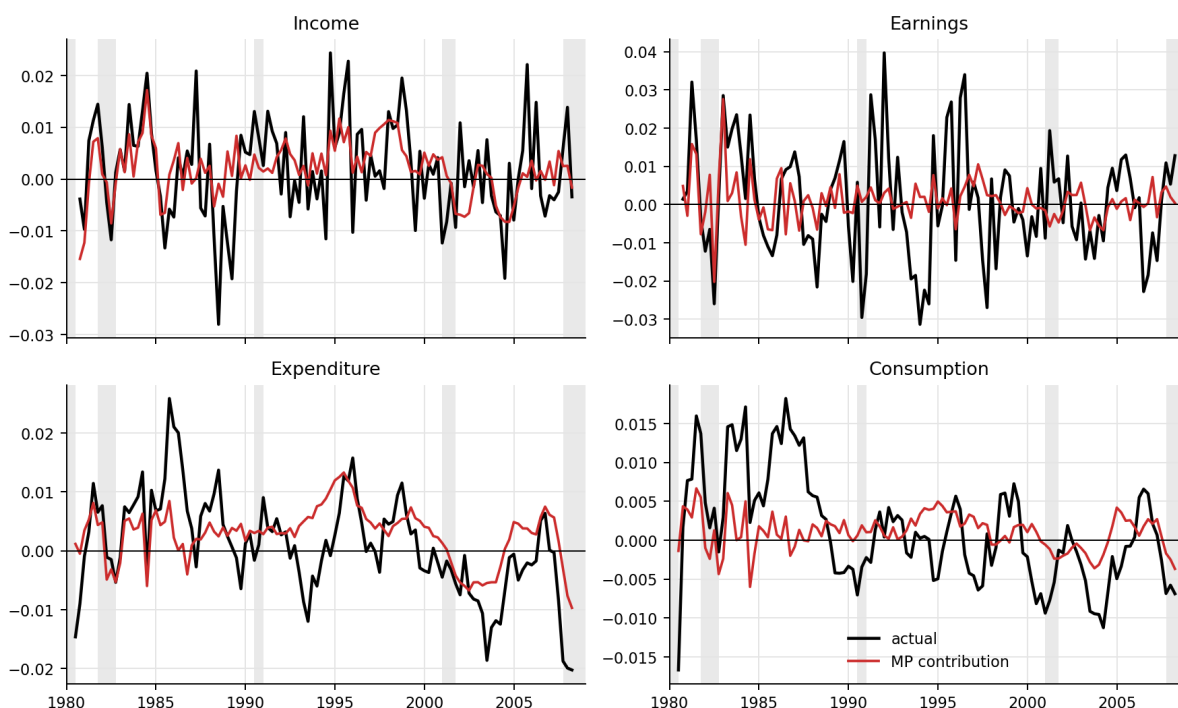


Figure 7: Predicted changes in inequality attributable to monetary policy shocks (thin line) versus actual observed changes (thick line), 1980–2008. Inequality measured by cross-sectional standard deviation. Series are three-quarter centered moving averages. Shaded areas: NBER recessions. Source: Replication of Figure 5 in [Coibion et al. \(2017\)](#).

The early-1980s Volcker disinflation stands out as the episode where monetary shocks most visibly account for inequality dynamics. In contrast, monetary shocks explain little of the sustained inequality trend after the mid-1980s, consistent with the paper’s point that identified shocks can only account for cyclical variation, not the long-run trend. The qualitative interpretation is fully replicated; axis magnitudes should be read with caution given the moving-average scaling issue documented in Section 5.

6.7 Inflation Target Shocks

Figure 8 replaces the RR shocks with the change in the Fed’s estimated implicit inflation target, measured following the time-varying Taylor rule of [Coibion and Gorodnichenko \(2011\)](#) and the structural model of [Ireland \(2006\)](#).

Figure 8: Effects of a 1 percentage point permanent decrease in the Fed’s inflation target on inequality (cross-sectional standard deviations). Top row: impulse responses with one and 1.65 SD confidence bands. Middle row: variance decompositions. Bottom row: historical contribution. Shaded areas: NBER recessions. Source: Replication of Figure 9 in [Coibion et al. \(2017\)](#).

All three rows of the figure are replicated with qualitatively indistinguishable results. A permanent 1 percentage point decrease in the inflation target raises all four inequality measures persistently. Changes in the inflation target account for a smaller share of forecast error variance than the broader RR shocks (consistently below 10%) but dominate the historical decomposition in the early 1980s, when the Volcker disinflation represented a large, permanent downward shift in the Fed’s inflation objective. The finding reinforces the paper’s conclusion that the Volcker disinflation was a particularly consequential episode for the evolution of US economic inequality.

7 Critical Assessment

7.1 The Exogeneity of Romer–Romer Shocks

The entire causal interpretation rests on the assumption that $\hat{\varepsilon}_m$ in (2) is orthogonal to all variables that independently affect inequality. This assumption is untestable in principle, but three practical threats have identifiable directional implications.

Omitted financial conditions. The Greenbook equation controls for output, inflation, and unemployment, but not for financial conditions: credit spreads, equity market volatility, or housing market indicators. If the Fed systematically responds to financial tightening that also independently raises inequality, because financial stress compresses business income at the top and depresses asset prices that support high-net-worth consumption, those responses end up in the residual. The bias runs upward: the residual picks up a positive association between tightening and inequality even absent a causal monetary policy effect. The high-frequency robustness check using federal funds futures ([Gorodnichenko and Weber, 2016](#)) partially addresses this, since intraday prices incorporate financial conditions, but it does not fully close the gap.

The information effect. [Nakamura and Steinsson \(2018\)](#) show that FOMC announcements convey the Fed’s private economic assessment beyond the policy decision

itself. When the Fed raises rates unexpectedly, markets partly interpret the move as a signal that the economy is performing better than publicly recognized, an optimistic signal embedded within the contractionary action. This good-economic-news component works *against* the inequality-raising direction of contractionary policy: a stronger-than-expected economy implies tighter labor market conditions, which disproportionately benefit lower-income workers through higher employment rates and wages. The identified Romer–Romer shock therefore captures a mixture of contractionary monetary policy (inequality-raising) and positive economic news (inequality-reducing via the labor market channel), and the estimated distributional effect would be *attenuated* toward zero relative to the pure policy effect. From this perspective, the CGKS estimates could be viewed as conservative: the true causal effect of a contractionary monetary policy shock, holding the economic outlook fixed, may be larger than the reported impulse responses suggest.

Reaction function stability. The original RR series covers 1969–1996; CGKS extend it to 2008 by re-estimating the same specification on the extended sample. This implicitly assumes the Fed’s reaction function was stable across three decades that span the Great Inflation, the Volcker disinflation, and the Greenspan tenure. If the systematic response to output shifted relative to the inflation response after the move toward implicit inflation targeting in the early 1990s, the post-1996 residuals may reflect a qualitatively different kind of variation than the original series. This is difficult to diagnose without the underlying FOMC-frequency data.

7.2 Sample Representativeness and the $\text{flag}=3$ Filter

As documented in Section 3, the consumption inequality series is built exclusively from Cohort A households. Cohort A households are simply those whose survey entry date happened to place their three spending months within the same calendar quarter. There is no obvious economic reason why they should behave differently from the excluded two-thirds. But the paper provides no evidence either way. The three-cohort robustness check in the appendix constrains the IRF to be equal across cohorts and finds qualitatively similar results, which is reassuring, but it is not a direct test of whether the cohorts differ.

7.3 The Top 1% Coverage Gap

The secular rise in US income inequality since the 1980s is driven predominantly by top-1% income growth (Piketty and Saez, 2003). Financial income, which rises significantly after contractionary shocks (visible in Figure 2), is concentrated at the very top of the distribution. The CEX misses these households entirely, through a combination of non-response and underreporting. The implication is clear: the income inequality estimates are lower bounds. If the financial income channel for top-1% households were captured, both the income inequality IRF and the variance decomposition contribution fractions would

be larger. The paper explicitly acknowledges this, but the magnitude of understatement is difficult to quantify without linking the CEX to administrative tax records.

7.4 Annual Income Timing

A less-discussed limitation concerns the temporal mismatch between income measurement and shock timing. A household interviewed in April 2000 reports income for April 1999 through March 2000; this annual figure is assigned to Q2 2000. The Q2 2000 monetary policy shock captures FOMC innovations in April, May, and June 2000. There is no temporal overlap between the income reference period and the shock.

The paper applies the contemporaneous shock specification (5) uniformly to all inequality series, including income. While this specification is justified for consumption (Section 4), it does not have an analogous timing justification for income: the Q2 2000 shock cannot logically affect income earned in the twelve months ending March 2000. This mismatch likely contributes both to the wider confidence bands for income inequality and to the characteristic delayed peak in the income IRF relative to consumption.

7.5 Sample Size and the Parameterization Problem

The estimation sample is approximately 115 quarterly observations. Each stacked local projection contains 483 slope terms plus 21 horizon fixed effects. This is heavily parameterized relative to the available data, and the wide confidence bands throughout Figure 3 are the direct consequence. The joint null rejection is credible because it accumulates evidence across all 21 horizons simultaneously. Specific quantitative claims (such as the 20% forecast error variance contribution of monetary policy to expenditure inequality) carry substantial sampling uncertainty and are further affected by the FEVD denominator inconsistency noted above.

7.6 The Zero-Lower-Bound Exclusion

The sample ends at 2008Q4, when the federal funds rate hit its effective lower bound. The distributional consequences of quantitative easing, forward guidance, and the near-zero rate environment of 2009–2015 are entirely outside the paper’s scope. The CGKS identification strategy cannot be extended to the ZLB period because the Greenbook-purged residual from a policy rate equation has no natural interpretation when the rate is constrained. Whether the directional finding (contractionary policy raises inequality) extends to asset-price-based unconventional policy, which operates primarily through wealth and portfolio effects at the top of the distribution, remains an open question.

8 Conclusion

The central finding of Coibion et al. (2017) holds up: contractionary monetary policy shocks persistently raise income, earnings, consumption, and expenditure inequality in the United States, and this is true across all twelve tested combinations of economic concept and inequality measure. The mechanism results (the asymmetric percentile responses, the net-worth group consumption gap, the Volcker-era dominance in the historical decomposition) come out with near-identical statistical conclusions to the original.

The replication also turns up some things worth noting. The pipeline that constructs the consumption inequality series silently drops two-thirds of CEX respondents through a sample selection step the paper does not describe; anyone building on this data construction should be aware of it. For income inequality, there is a timing mismatch the paper papers over: annual income reference periods are assigned to the quarter of measurement rather than receipt, which sits awkwardly with a contemporaneous shock specification applied uniformly to all twelve series. The historical decomposition figure has a denominator error in the moving average that inflates the y-axis, and the FEVD formula likely overstates long-horizon contribution fractions. None of this changes the qualitative conclusions.

The methodologically most important feature of the paper is the stacked local projection with Driscoll-Kraay standard errors and the joint test across all 21 horizons. Individual-horizon impulse responses are imprecise, as the wide confidence bands make clear, but testing whether the entire response path is jointly zero is the right question, and every series rejects it. The sign is consistent across all twelve inequality measures over a five-year horizon, which is ultimately what makes the finding credible.

The CGKS impulse responses sit between the predictions of Gornemann et al. (2016), who predict roughly one-third of the estimated income Gini response, and Luetticke (2015), who predict roughly twice as large an effect. Standard HANK calibrations span the empirically relevant range, but none consistently matches the magnitude. Extending the identification strategy to the unconventional monetary policy era, where asset price channels dominate, is the natural next step.

The replication code is available at <https://github.com/omerfyldz/Inequality-caused-by-mone>

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